

Notes: Diamond & Dybvig (JPE, 1983)

Bank Runs, Deposit Insurance, and Liquidity

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1 Big picture

- **Question.** Why do banks exist, why are they fragile, and why can deposit insurance improve welfare?
- **Core friction.** Agents face privately observed liquidity needs. Some agents need to consume early, while others can wait. Because the type is private information, simple markets cannot write direct insurance contracts contingent on liquidity type.
- **Role of banks.** A bank invests in illiquid long-term assets and issues liquid demand deposits. This transforms illiquid assets into liquid liabilities and improves risk sharing relative to direct holding of assets.
- **Fragility.** The same demand-deposit contract that provides liquidity also creates multiple equilibria. One equilibrium is efficient: only impatient depositors withdraw early. Another equilibrium is a run: all depositors rush to withdraw because they expect others to do so.
- **Real cost of runs.** Runs are not merely accounting events. They force premature liquidation of long-term production and destroy risk sharing.
- **Suspension of convertibility.** If the number of impatient depositors is known, a bank can stop withdrawals after the right threshold. This prevents runs and implements the efficient allocation.
- **Limitation of suspension.** If the number of impatient depositors is stochastic and unobservable, suspension cannot implement the full-information optimum under sequential service.
- **Deposit insurance.** Government deposit insurance, financed by taxes, can separate asset liquidation from withdrawal volume. It rules out runs and can implement the unconstrained optimum.
- **Policy message.** The central policy problem is not only solvency. It is coordination. A credible guarantee can eliminate the bad equilibrium before it occurs.
- **Broader implication.** Any institution with liabilities more liquid than assets can face run-like coordination failures, including firms reliant on short-term debt.

2 Data and measurement: the theoretical environment

This is a theoretical paper. There are no empirical tables or figures in the original article. The “measurement” objects are the model’s primitive variables: time, technology, preference types, withdrawal fractions, and feasible contracts.

2.1 Three-period economy

- There are three dates: $T = 0, 1, 2$.
- There is a single homogeneous good.
- Each agent is endowed with one unit of the good at $T = 0$ and nothing later.
- Long-term technology transforms one unit invested at $T = 0$ into $R > 1$ units at $T = 2$.
- If the technology is liquidated early at $T = 1$, it returns only one unit.

$$\text{Invest 1 at } T = 0 \quad \Rightarrow \quad \begin{cases} 1 & \text{if liquidated at } T = 1, \\ R & \text{if held until } T = 2. \end{cases}$$

Interpretation. Illiquidity is modeled as a maturity mismatch. The project is productive if held to maturity but costly to liquidate early.

2.2 Private liquidity types

- At $T = 1$, each agent privately learns whether he is type 1 or type 2.
- Type 1 agents are impatient and care only about consumption at $T = 1$.

- Type 2 agents are patient and care only about consumption at $T = 2$.
- The type is private information and cannot be directly contracted on.
- A fraction $t \in (0, 1)$ of agents are type 1.

$$U(c_1, c_2; \Theta) = \begin{cases} u(c_1), & \text{type 1,} \\ \rho u(c_1 + c_2), & \text{type 2.} \end{cases}$$

Interpretation. The demand for liquidity is an insurance demand: before learning their type, agents want protection against becoming impatient.

2.3 Assumptions on preferences

- u is increasing and strictly concave.
- Inada conditions hold: $u'(0) = \infty$ and $u'(\infty) = 0$.
- The discount factor satisfies $1 \geq \rho > R^{-1}$.
- Relative risk aversion is greater than one:

$$-\frac{cu''(c)}{u'(c)} > 1.$$

Interpretation. These assumptions imply a strong desire to insure impatient agents. In the optimum, impatient agents consume more than they would by self-liquidating, while patient agents receive less than the full long-run return.

2.4 Sequential service

- Demand deposits are paid sequentially: the bank pays depositors in the order they arrive.
- A depositor's payoff at $T = 1$ can depend on his place in line, but not on how many depositors will arrive later.
- This is the key constraint that makes private bank contracts fragile.

Interpretation. Sequential service captures the real-world feature that withdrawals are honored as they arrive. It prevents the bank from perfectly conditioning early payments on the total number of true liquidity shocks.

3 Models and theoretical specifications

3.1 Competitive market without banks

If agents hold the long asset directly, type 1 agents liquidate at $T = 1$ and receive

$$c_1^1 = 1,$$

while type 2 agents wait until $T = 2$ and receive

$$c_2^2 = R.$$

Interpretation. Direct asset holding provides no insurance against becoming impatient. It is technologically feasible, but not ex ante risk-sharing efficient.

3.2 Full-information optimal risk sharing

If types were publicly observable, the optimal allocation satisfies

$$c_1^{2*} = c_2^{1*} = 0, \quad (1a)$$

$$u'(c_1^{1*}) = \rho R u'(c_2^{2*}), \quad (1b)$$

$$tc_1^{1*} + \left[(1-t) \frac{c_2^{2*}}{R} \right] = 1. \quad (1c)$$

- Equation (1a) says impatient agents consume only at date 1 and patient agents consume only at date 2.
- Equation (1b) equates marginal utility adjusted by the technological return and discounting.
- Equation (1c) is the resource constraint.

Under the paper's assumptions:

$$c_1^{1*} > 1, \quad c_2^{2*} < R, \quad c_2^{2*} > c_1^{1*}.$$

Interpretation. The full-information optimum transfers resources toward type 1 agents relative to direct holding. This is exactly the liquidity insurance that banks can provide.

3.3 Demand deposit contract

A demand deposit promises r_1 per unit deposited to anyone who withdraws at $T = 1$, subject to available assets. Let f_j be the fraction of deposits served before depositor j , and let f be the total fraction of deposits withdrawn at $T = 1$.

The date-1 payoff is

$$V_1(f_j, r_1) = \begin{cases} r_1, & f_j < r_1^{-1}, \\ 0, & f_j \geq r_1^{-1}. \end{cases}$$

The date-2 payoff is

$$V_2(f, r_1) = \max \left\{ \frac{R(1 - r_1 f)}{1 - f}, 0 \right\}.$$

Interpretation. If $r_1 = c_1^{1*}$ and only type 1 agents withdraw, the demand deposit implements the full-information optimum. The same contract also has a run equilibrium.

3.4 The good equilibrium

If all agents expect type 1 agents to withdraw and type 2 agents to wait:

$$f = t.$$

Then:

$$V_1 = c_1^{1*}, \quad V_2 = c_2^{2*}.$$

Interpretation. The bank uses early liquidation to pay impatient agents and leaves enough assets invested for patient agents. This is efficient liquidity transformation.

3.5 The run equilibrium

If everyone expects everyone else to withdraw at $T = 1$, then it is optimal for every depositor to withdraw at $T = 1$. The bank liquidates its long assets early, destroying the insurance arrangement.

Interpretation. The run is self-fulfilling. It need not be caused by fundamental insolvency; it can be triggered by expectations, public signals, or sunspots.

3.6 Suspension of convertibility

Under suspension, withdrawals stop after a threshold $\hat{f}$. The modified payoff is

$$V_1(f_j, r_1) = \begin{cases} r_1, & f_j \leq \hat{f}, \\ 0, & f_j > \hat{f}, \end{cases}$$

and

$$V_2(f, r_1) = \max \left\{ \frac{(1 - fr_1)R}{1 - f}, \frac{(1 - \hat{f}r_1)R}{1 - \hat{f}} \right\}.$$

If t is known and $\hat{f}$ is chosen appropriately, type 2 agents never want to withdraw early.

Interpretation. Suspension works because it removes the first-mover advantage in a run. However, it works cleanly only when the bank knows the normal withdrawal volume.

3.7 Stochastic withdrawals and Proposition 1

When the fraction of impatient agents t is stochastic, the bank does not know how many legitimate withdrawals will occur. The paper proves:

Proposition 1. Bank contracts obeying sequential service cannot achieve optimal risk sharing when t is stochastic and nondegenerate.

Interpretation. The bank faces a trade-off. If it pays early withdrawers a constant amount, consumption cannot adjust optimally to the realized number of type 1 agents. If it pays different amounts depending on place in line, type 1 agents receive unequal consumption. Both violate the unconstrained optimum.

3.8 Deposit insurance and Proposition 2

Government deposit insurance guarantees the promised withdrawal value and finances any shortfall through taxes. The key difference from a bank is that the government can tax after observing total withdrawals.

The paper's tax function is

$$\tau(f) = \begin{cases} 1 - \frac{c_1^{1*}(f)}{r_1}, & f \leq \bar{t}, \\ 1 - r_1^{-1}, & f > \bar{t}, \end{cases}$$

where $\bar{t}$ is the largest possible realization of the fraction of type 1 agents.

The after-tax early payoff is

$$\hat{V}_1(f) = \begin{cases} c_1^{1*}(f), & f \leq \bar{t}, \\ 1, & f > \bar{t}. \end{cases}$$

The after-tax late payoff is

$$\hat{V}_2(f) = \begin{cases} \frac{R\{1 - [c_1^{1*}(f)f]\}}{1 - f} = c_2^{2*}(f), & f \leq \bar{t}, \\ R, & f > \bar{t}. \end{cases}$$

Proposition 2. Demand deposits with government deposit insurance achieve the unconstrained optimum as a unique dominant-strategy equilibrium.

Interpretation. Deposit insurance breaks the link between withdrawal order and asset liquidation. Because type 2 agents know they will be paid if they wait, they do not join a run.

4 Conceptual design and interpretation

4.1 What identifies the role of banks in the model

- The model compares three allocations: direct asset holding, full-information risk sharing, and bank contracts.
- Direct holding is feasible but lacks liquidity insurance.
- Full-information risk sharing is the first-best benchmark but cannot be implemented by simple markets because types are private.
- Demand deposits implement the first-best in one equilibrium but also create a bad equilibrium.

Interpretation. Banks exist because they can implement liquidity insurance that markets cannot. They are fragile because the same contract relies on confidence.

4.2 Why runs are real, not just redistributive

- In a run, long-term assets are liquidated at date 1.
- Liquidation prevents the high return R from being realized.
- Type 2 agents lose the benefit of waiting until date 2.
- The economy loses the risk-sharing arrangement and the productive return.

Interpretation. A run is inefficient even if the bank's assets are fundamentally safe. The coordination failure itself creates real damage.

4.3 Why deposit insurance is powerful

- Deposit insurance changes expectations.
- It removes the incentive for patient depositors to rush.
- If the guarantee is credible, the bad equilibrium disappears.
- In the deterministic-withdrawal case, the guarantee can be costless in equilibrium because it never needs to be used.

Interpretation. The policy is not mainly about bailing out failed banks after the fact. It is about preventing panic before liquidation occurs.

4.4 Limits of the model

- The technology is riskless; real bank assets are risky.
- The model abstracts from moral hazard in bank portfolio choice.
- It analyzes one representative bank or banking system, not interbank liquidity markets.
- It treats deposit insurance as tax-financed and credible.
- It does not model the optimal design of regulation that should accompany insurance.

Interpretation. Later banking theory extends Diamond–Dybvig by adding asset risk, moral hazard, capital requirements, interbank markets, and lender-of-last-resort discretion.

5 Extracted equation panels from the original paper

The original article contains no empirical tables or plotted figures. To preserve the “direct extraction” format, the notes below include the key equation panels and proposition blocks cropped from the source paper.

5.1 Technology and preferences

Read:

- The production technology is long-term and illiquid: early liquidation gives 1, waiting gives $R > 1$.
- Utility is type dependent: impatient type 1 agents value c_1 , patient type 2 agents value $c_1 + c_2$ at date 2.

Economic message. The model creates liquidity demand from private information. Agents want insurance before they know whether they will become impatient.

$$\begin{array}{ccc} T = 0 & T = 1 & T = 2 \\ -1 & \begin{cases} 0 \\ 1 \end{cases} & \begin{matrix} R \\ 0 \end{matrix} \end{array}$$

(a) Original technology block.

$$U(c_1, c_2; \Theta) = \begin{cases} u(c_1) & \text{if } j \text{ is of type 1 in state } \Theta \\ \rho u(c_1 + c_2) & \text{if } j \text{ is of type 2 in state } \Theta, \end{cases}$$

(b) Original utility block.

5.2 Full-information optimum and demand deposits

Read:

- The first-best allocation delays type 2 consumption and transfers some resources to type 1 agents.
- Demand deposits can implement the first-best in the good equilibrium.
- The same demand deposit contract also creates a run equilibrium.

Economic message. The banking contract is both useful and fragile because it promises early liquidity backed by illiquid long-term assets.

$$c_1 = c_2 = 0$$

(a) Equation (1a).

(1a)

$$\begin{aligned} u'(c_1^*) &= \rho R u'(c_2^*) \\ &\text{1 line with marginal productivity), and} \\ t c_1^* + [(1 - t) c_2^* / R] &= 1 \end{aligned}$$

(b) Equations (1b)–(1c).

$$V_1(f, r_1) = \begin{cases} r_1 & \text{if } f_j < r_1^{-1} \\ 0 & \text{if } f_j \geq r_1^{-1} \end{cases} \quad (2)$$

(1b)

$$V_2(f, r_1) = \max \{R(1 - r_1 f) / (1 - f), 0\}, \quad (3)$$

(1c)

the number of withdrawers' deposits serviced before agent

(c) Demand-deposit payoffs.

5.3 Suspension and stochastic withdrawals

Read:

- Suspension stops withdrawals after a threshold $\hat{f}$.
- With deterministic t , this can prevent a run without actually binding in the good equilibrium.
- With stochastic t , Proposition 1 says sequential-service bank contracts cannot implement the full-information optimum.

Economic message. Suspension helps with runs, but it cannot perfectly separate true liquidity needs from panic when the normal withdrawal volume is uncertain.

$$V_1(f, r_1) = \begin{cases} r_1 & \text{if } f_j \leq \hat{f} \\ 0 & \text{if } f_j > \hat{f} \end{cases}$$

$$V_2(f, r_1) = \max \left\{ \frac{(1 - fr_1)R}{1 - \frac{f}{\hat{f}}}, \frac{(1 - \hat{f}r_1)R}{1 - \frac{\hat{f}}{\hat{f}}} \right\},$$

(a) Suspension-of-convertibility payoffs.

Optimal risk sharing.

PROPOSITION 1: Bank contracts (which must obey the sequential service constraint) cannot achieve optimal risk sharing when t is stochastic and has a nondegenerate distribution.

Proposition 1 holds for all equilibria of uninsured bank contracts of

(b) Proposition 1 block.

5.4 Deposit insurance

Read:

- Deposit insurance uses taxation to guarantee withdrawal payments.
- The tax function can be based on total withdrawals f , which banks cannot know while serving depositors sequentially.
- The after-tax payoffs restore the full-information allocation.

Economic message. Insurance eliminates the run incentive by making waiting safe for patient agents. The policy separates liquidity provision from forced asset liquidation.

$$\tau(f) = \begin{cases} 1 - \frac{c_1^*(f)}{r_1} & \text{if } f \leq \bar{t} \\ 1 - r_1^{-1} & \text{if } f > \bar{t}, \end{cases}$$

test possible realization of $\bar{t}$

(a) Tax function.

$$\hat{V}_1(f) = \begin{cases} c_1^*(f) & \text{if } f \leq \bar{t} \\ 1 & \text{if } f > \bar{t}. \end{cases}$$

(b) After-tax early payoff.

$$\hat{V}_2(f) = \begin{cases} \frac{R\{1 - [c_1^*(f)f]\}}{1 - f} = c_2^{**}(f) & \text{if } f \leq \bar{t} \\ \frac{R(1 - f)}{1 - f} = R & \text{if } f > \bar{t}. \end{cases}$$

(c) After-tax late payoff.

6 Short summary

- The paper explains why banks exist and why they are fragile.
- Agents face private liquidity shocks, making direct insurance contracts infeasible.
- Long-term assets are productive but illiquid.
- Demand deposits provide liquidity insurance by transforming illiquid assets into liquid claims.
- The good equilibrium implements efficient risk sharing: impatient agents withdraw, patient agents wait.
- The run equilibrium is self-fulfilling: patient agents also withdraw because they expect others to withdraw.
- Runs cause real losses by forcing liquidation and destroying risk sharing.
- Suspension of convertibility can prevent runs when the normal withdrawal volume is known.
- When withdrawals are stochastic, sequential-service constraints prevent private bank contracts from achieving the full-information optimum.
- Government deposit insurance can rule out runs and implement the unconstrained optimum by using taxation to guarantee payments.
- The model justifies deposit insurance as a way to eliminate a bad equilibrium, not merely as a transfer after failure.
- The paper also suggests that run-like fragility applies beyond banks to any institution funded by short-term liquid claims and invested in illiquid assets.

7 Conclusion

7.1 Core conclusion

Diamond and Dybvig show that banks are socially valuable because they provide liquidity insurance that simple markets cannot provide under private information. However, the same demand-deposit contract that supplies liquidity also creates the possibility of self-fulfilling runs. Bank fragility is therefore not an accidental flaw; it is tied to the liquidity transformation function of banks.

7.2 Economic intuition

The intuition is that impatient consumers need early consumption, while patient consumers benefit from waiting for high long-run returns. A bank can pool liquidity risk and pay impatient consumers early while investing the rest long term. But if patient consumers fear that others will withdraw, they also rush to withdraw. The bank must then liquidate long-term assets early, validating the panic.

7.3 Takeaway

The enduring lesson is that financial stability is a coordination problem. A credible guarantee, such as deposit insurance, can remove the bad equilibrium and preserve productive investment. The model remains foundational because it links liquidity creation, maturity transformation, runs, suspension, deposit insurance, and lender-of-last-resort policy in one simple framework.